

Lecture 4. Random Algebraic Construction

- KST : $ex(n, K_{s,t}) = O(n^{2-1/s})$, $2 \leq s \leq t$.

$t \gg s$ (norm graphs)

Today { • Bukh : $\Omega(n^{2-1/s})$ for $K_{s,t}$ -free graphs
 $t \gg s$.
• Bukh-Conlon : rational exponent conj.

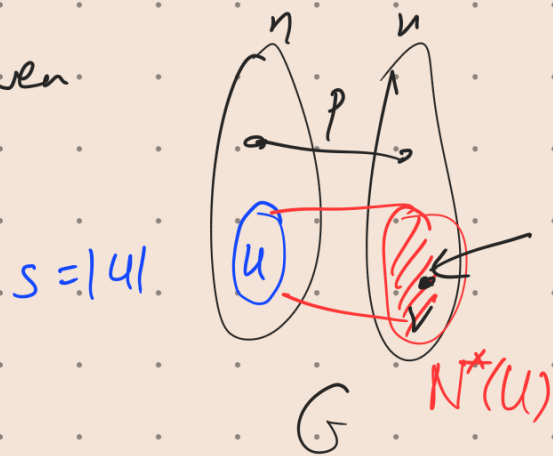
Notation $U \subseteq V(G)$, $N^*(U) =$ common neighbors of U
 $= \bigcap_{u \in U} N(u)$

1. Long smooth tail in $G(n, p)$ construction.

- Build (n, n) -bip, edges given

w./ prob. $p = n^{-1/s}$

$$\mathbb{E} e(G) = n^2 p = n^{2-1/s}$$



- WTS, for some t , $\Pr(K_{s,t} \subseteq G) \rightarrow 0$

Fix an s -set U , $\Pr(v \in N^*(U)) = p^{|U|} = n^{-1}$.

r.v. $|N^*(U)| \sim \text{Binom}(n, 1/n)$

$\sim$ Poisson with mean 1.

$$\Rightarrow \Pr(|N^*(u)| \geq t) \leq \frac{1}{t!}$$

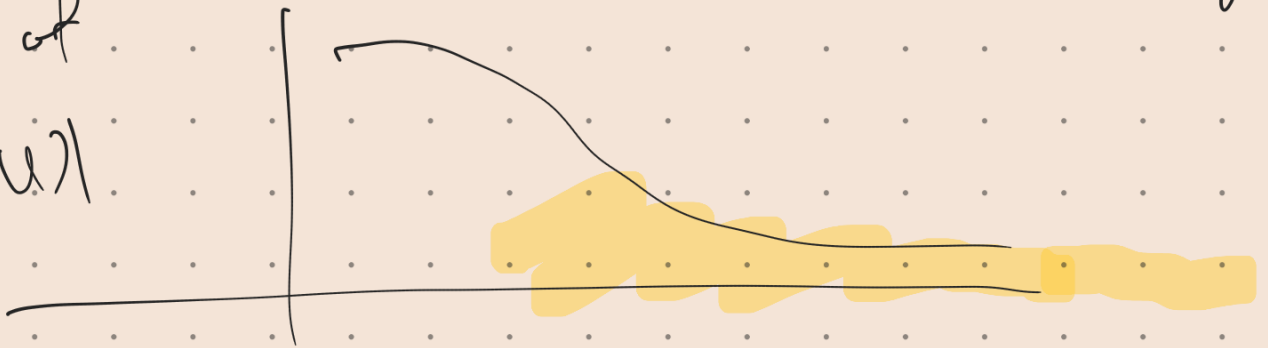
• Union bound

$$\Rightarrow \Pr(K_{s,t} \leq G) \leq 2 \cdot \binom{n}{s} \cdot \frac{1}{t!} \rightarrow 0$$

$$\text{if } t \geq 10s \frac{\log n}{\log \log n}$$

• Known G whp contains $K_{s,t'}$; $t' = \frac{s}{10} \frac{\log n}{\log \log n}$

dist. of $|N^*(u)|$



2. Random algebraic construction.

Feature { locally enjoy randomness

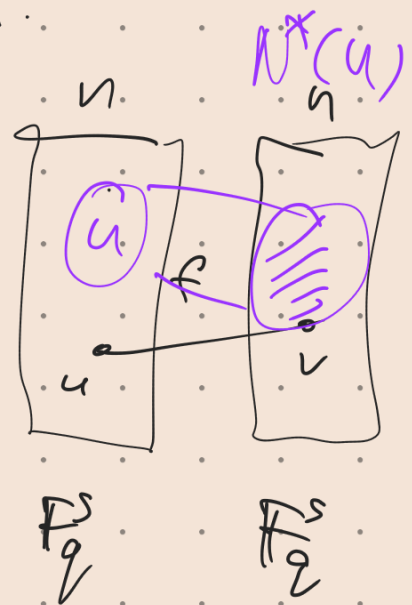
$N^*(u)$

$\hookrightarrow$ alg. var.

$|N^*(u)|$

possible values

have discontinuity.



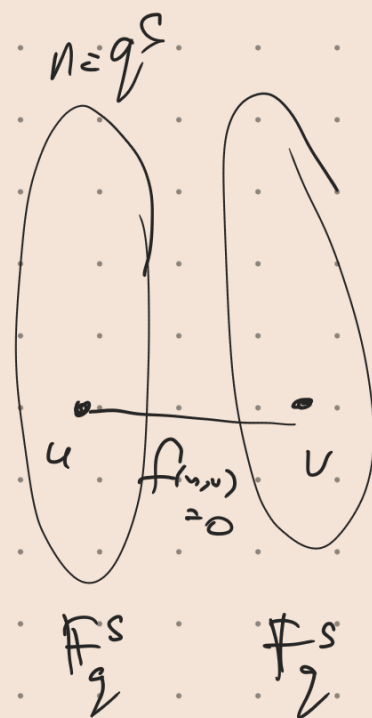
Construction • Goal (n, n) w. $\sim \Theta(n^{2-1/s})$ edges
 $K_{s,t}$ -free

Let $s \geq 4$, $d = s^2 - s + 2$, q large prime power

$$n = q^s.$$

• Take $f: \mathbb{F}_q^s \times \mathbb{F}_q^s \rightarrow \mathbb{F}_q$

a unif random polynomial
 on $2s$ -variables over $\mathbb{F}_q$
 with $\deg \leq d$.



• $u \sim_G v \Leftrightarrow f(u, v) = 0$

Lem 2.1 Let f be as above. Then $\forall u, v \in \mathbb{F}_q^s$

$$\Pr(f(u, v) = 0) = 1/q$$

$$\bullet \mathbb{E} e(G) = n^2/q = n^{2-1/s}$$

concentration $\Rightarrow$ whp $e(G) \geq \frac{1}{2} n^{2-1/s}$

Pf: • Let $c \in \mathbb{F}_q$ be the constant term in f .

Set $g = f - c$

f unif $\rightsquigarrow$ c unif dist. in $\mathbb{F}_q$

• Condition on the value of $g(u,v)$,

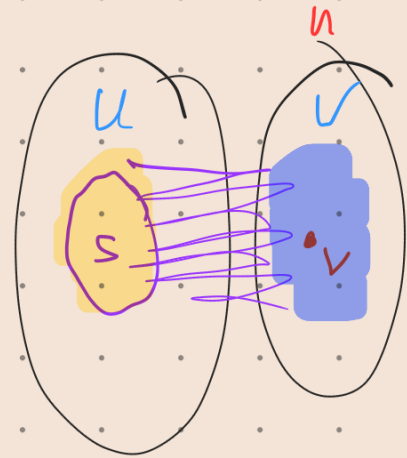
$$f(u,v)=0 \Leftrightarrow c = -g(u,v) \in \mathbb{F}_q$$

which happens w./ prob. $1/q$.



Lemma 2.2 Let f as before, Let $U, V \subseteq \mathbb{F}_q^s$ of size s, r (resp.) with $s, r \leq \min(\sqrt{q}, d)$

$$\Rightarrow \Pr(f(u,v)=0, \forall u \in U, \forall v \in V)$$



$$= q^{-sr}$$

$$\mathbb{E}|N^*(u)|^2 = n \cdot \frac{1}{n} = 1$$

Now we proceed to bound $\mathbb{E}(\#K_{s,t})$, t large.

Say s -set U in a partite is **bad** if

$$|N^*(U)| \geq t, \text{ NTS } \# \text{ bad } s\text{-sets} = \text{few}$$

Fix an s -set U ,

We shall bound the moments of

the r.v. $|N^*(U)|$.

• Note that

$$|N^*(u)| = \sum_{v \in \mathbb{F}_q^S} \mathbb{1}_{N^*(u)}(v)$$



$\Rightarrow$ d^{th} -moment

$$\mathbb{E}(|N^*(u)|^d) = \mathbb{E}\left(\left(\sum_{v \in \mathbb{F}_q^S} \mathbb{1}_{N^*(u)}(v)\right)^d\right)$$

$$= \mathbb{E}\left[\sum_{v_1, \dots, v_d \in \mathbb{F}_q^S} \mathbb{1}_{N^*(u)}(v_1) \cdots \mathbb{1}_{N^*(u)}(v_d)\right]$$

$$= \sum_{v_1, \dots, v_d \in \mathbb{F}_q^S} \mathbb{E}\left(\mathbb{1}_{N^*(u)}(v_1) \cdots \mathbb{1}_{N^*(u)}(v_d)\right)$$

if $\{v_1, \dots, v_d\}$ has $r \leq d$ distinct vertices

Lemma 2.2

$$\Rightarrow \mathbb{E}(\cdot) = q^{-sr}$$



$$\Rightarrow = \sum_{r \leq d} \binom{q^S}{r} \cdot M_r \cdot q^{-sr} \leq \sum_{r \leq d} M_r = O_d(1)$$

where $M_r = \# \text{ surj functions from } [d] \text{ to } [r]$

$$\mathbb{E}(|N^*(u)|^d) = O_d(1)$$

View $N^*(u)$ as an alg. var.

$$N^*(u) = \{v \in \mathbb{F}_q^s : f(u, v) = 0 \forall u \in \mathcal{U}\}$$

$$= \bigcap_{u \in \mathcal{U}} \{f(u, \cdot) = 0\}$$



The Lang-Vojta bound:

$$\begin{cases} \text{either} & |N^*(u)| \leq C \\ \text{or} & |N^*(u)| \geq q/2 \end{cases}$$

$$(C = O(1) \text{ as } q \rightarrow \infty)$$

$$\Pr(|N^*(u)| > C) = \Pr(|N^*(u)| \geq q/2)$$

$$= \Pr(|N^*(u)|^d \geq (q/2)^d) \stackrel{\text{Markov}}{\leq} \frac{\mathbb{E}|N^*(u)|^d}{(q/2)^d}$$

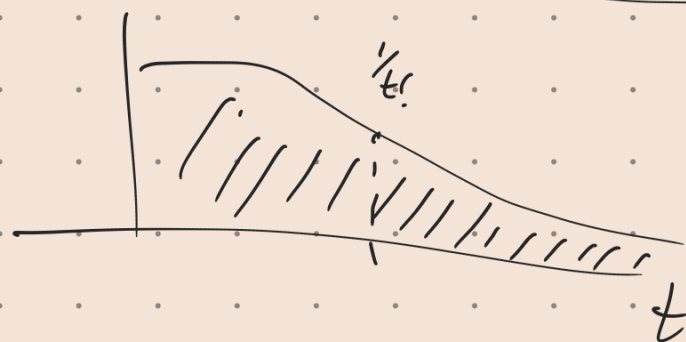
$$= \frac{O_d(1)}{q^d}$$

$$\Rightarrow \mathbb{E}(\# \text{ bad } s\text{-set}) \leq 2 \binom{n}{s} \frac{O_d(1)}{q^d} = O(q^{s-2})$$

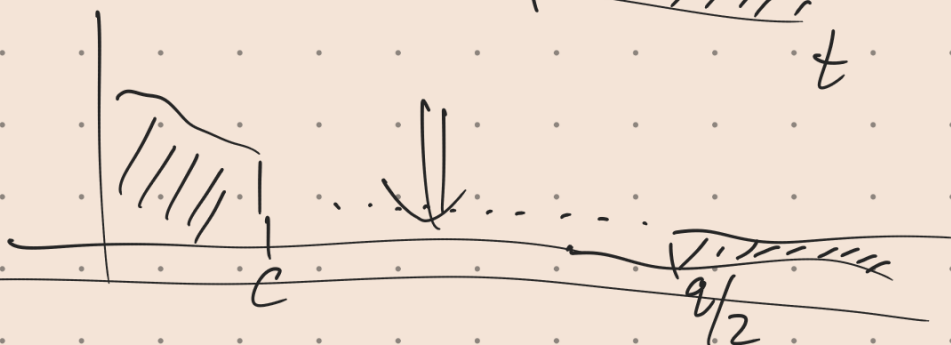
- Delete a u_x from each of the bad s -set $\Rightarrow$ No bad s -set left. 

$\Theta(n, p)$

$|N^*(u)|$



Now
Lang Weir



3. Rational Turán Exponent.

Def $r \in \mathbb{Q}$, $1 \leq r \leq 2$, is Turán exponent if $\exists H$ s.t. $ex(n, H) = \Theta(n^r)$

Conj (RTE) $\forall$ rational $r \in (1, 2)$

it is a Turán exponent.

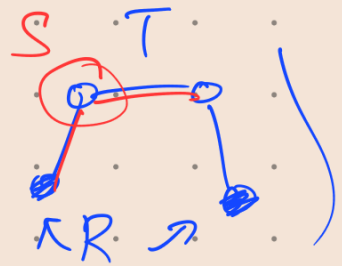
(B-C) $\forall$ $r \in (1, 2)$ rational, $\exists$ finite family of graphs $\mathcal{H}$ s.t.
 $ex(n, \mathcal{H}) = \Theta(n^r)$

- It comes from blowups of rooted trees.

Def A rooted tree (T, R) $\left\{ \begin{array}{l} \text{tree } T \\ \text{roots } R \\ \text{(indep set)} \end{array} \right.$

$$\text{density} = \rho_T = \frac{e(T)}{|T| - |R|}$$

$$(\rho_T = \frac{3}{2})$$



Call (T, R) balanced if $\forall S \subseteq V(T) \setminus R$

unrooted vxs, $\rho_S \geq \rho_T$, where

$$\rho_S = \frac{e(S)}{|S|}$$

$$\rho_S = \frac{2}{1} = 2 \geq \frac{3}{2} = \rho_T$$

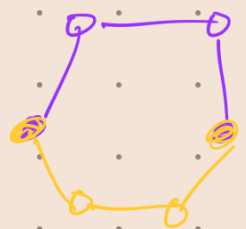
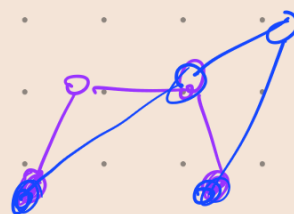
p th power of (T, R) $\mathcal{T}^p$ family of graphs

- taking p copies of (T, R)

- identify them at R

- no restriction to unrooted vxs.

$p=2$



- Roughly speaking, balanced $\approx$ ^{rooted} leaves are "evenly distributed"



Lem 3.4 Let (T, R) be a rooted tree w/ $R \neq \emptyset$

$$\Rightarrow ex(n, T^p) = O_p(n^{2 - \frac{1}{p_T}})$$

Lem 3.5 Let (T, R) be a balanced rooted tree,

then $\exists p \in \mathbb{N}$ s.t. $ex(n, T^p) = \Omega_p(n^{2 - \frac{1}{p_T}}) \dots (X)$

Ex 3.5 Find an example of unbalanced rooted tree s.t. the lower bound (X) does NOT hold.

• Upp bd Lem 3.4 follows by averaging.

• Lower bd Lem 3.6.

• a unrooted

Let (T, R) be rooted tree w./

• $r = |R|$ roots

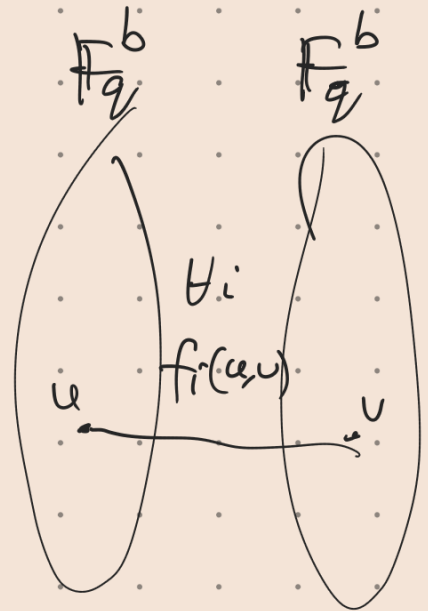
• $b = \text{edges}$

$$\rho_T = \frac{b}{a} \quad \Omega(n^{2 - \frac{1}{\rho_T}}) = \Omega(n^{2 - \frac{a}{b}})$$

• $S = 2br$, $d = sb$, $n = q^b$

Take $f_1, \dots, f_a$ indep. random polyn.

over $\mathbb{F}_q$ w./ $\begin{cases} \deg \leq d \\ \# \text{var. } 2b \end{cases}$



$$u \sim_{\mathbb{F}^V} v \Leftrightarrow f_i(u, v) = 0 \quad \forall i \in [a]$$

$$\mathbb{E}_e(G) = q^{2b} \cdot q^{-a} = n^{2 - \frac{a}{b}} \quad \checkmark$$

NTS few copies (in expectation) of graphs from $\mathcal{GP}$.

Fix $\{w_1, \dots, w_r\} = W$, let $\mathcal{U} =$ collection of all copies of T in G rooted at W .

Now bound the moments of $|\mathcal{U}|$.

Before

$$|N^*(u)|$$



$$N^*(u)$$

root

bad

if $|N|$ large

u



u

